

Maestro

Production Specification Package

DOCUMENT	v6 Production Spec – 7-Deliverable Package
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"Every screen must answer a decision-making question. If a screen exists because the engine has this data, it is infrastructure. If it exists because a leader would change a decision after seeing this, it is product."

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1. Product Specification (PRD)

This Product Requirements Document defines Maestro v6 — the Organizational Judgment Infrastructure product. It supersedes the v1 "browser-first Agent Operating System" prototype, which has been retired. The product is positioned as a Bloomberg-terminal equivalent for organizational execution: customers buy confidence, reduced uncertainty, faster decisions, fewer mistakes, institutional memory, and organizational intelligence — not infrastructure, not dashboards, not AI chat.

1.1 Vision

Maestro is the system of record for organizational judgment. It reconstructs how a company executes work by ingesting signals from Jira, GitHub, Slack, Confluence, and other systems; infers the organizational laws that explain why the company behaves the way it does; and surfaces decisions, predictions, and dissent to executives in a form that changes what they decide. The success criterion is binary and weekly: did Maestro cause the organization to make a decision it would not otherwise have made?

Rule Zero — Frozen until 3 design partners use the OEM and make decisions because of it. No new core abstractions. No new surfaces. No new metrics. Ship the wedge.

1.2 The 5-Layer Stack

Maestro is architected as five layers. The bottom four describe how the product produces answers; the fifth describes how the product changes what the customer believes. This separation is intentional — the Belief Layer is where Maestro stops being a tool and starts being a trusted cognition partner.

Layer	Name	Purpose
L5	Belief Layer	Calibration, trust, Surprise Hit Rate (SHR) — the only mechanism by which trust compounds over time
L4	Executive Cognition	CEO-facing product: Inbox, Decisions, Physics, Debate, Live Meeting Intelligence
L3	Decision Engine	Recommendations, counterfactual simulation, prediction logging
L2	Organizational Execution Model (OEM)	The living model of the company: entities, relations, laws, capabilities
L1	Signals	Raw observations from GitHub, Jira, Slack, Confluence, Figma, plus external (LinkedIn, Twitter, News)

1.3 Personas

Maestro has one primary persona and three secondary personas. The primary persona is the CEO; the product fails if the CEO does not use it daily. Secondary personas exist to support the CEO's decision-making, not as independent users. This is a critical design constraint — features that serve secondary personas at the expense of the CEO are explicitly forbidden during the Rule Zero window.

Persona	Role	Primary Verb	Daily Session
Jane (CEO)	Fortune 500 CEO	Decide	30 min morning + 5 ad-hoc
Chris (CTO)	Engineering leader	Recommend	15 min + 2 meetings
Casey (CFO)	Finance leader	Validate	10 min during decision review
Pat (VP Sales)	Revenue leader	Dissent	As-needed during debates

1.4 Functional Requirements

Onboarding — Time-to-First-Insight (TTFS)

Onboarding must produce a "Wait... really?" moment within 31 days, structured as a 6-reveal sequence. Each reveal corresponds to a model-readiness milestone, not a fake real-time countdown. Honesty about model maturity is a v6 design principle — pretending the model is ready before it is destroys trust permanently.

Reveal	Timing	Content	Model Readiness
R0	Minute 0	Welcome + signal connection	Pre-scan
R1	Hour 48	Duplicated work detection (OAuth example)	Structural patterns
R2	Hour 52	Causal bottleneck (Legal enters too late)	Causal patterns
R3	Day 14	Undocumented influence (Priya deploy expert)	Influence patterns
R4	Week 3	Counterfactual prediction (Platform +2 engineers)	Counterfactuals
R5	Day 31	19 organizational laws, 4 unknown to leadership	Laws inferred

Surfaces — Six Decision-Making Screens

v6 ships exactly six surfaces. Every surface must declare a Decision Question (DQ) — a one-sentence question the surface answers. Surfaces without a DQ are forbidden by design rule and rejected by the validateDecisionQuestion() function at runtime.

Surface	Decision Question (DQ)
Home	What must I decide today, and what is emerging?
Inbox	What needs my judgment, not my team's?
Decisions	What will happen if I approve this?
Physics	What explains why my company behaves this way?
Debate	Where does my org disagree that I haven't ruled on?
Live	What is being decided in this meeting, and what should I say next?

Live Meeting Intelligence

Live Meeting Intelligence is the v6 implementation of meeting-assistant functionality. It is explicitly NOT a Cluely clone. Three design decisions differentiate it: (1) transparency — every participant consents and can see the panel; (2) OEM connection — every meeting becomes a signal event that updates the organizational model; (3) synthesis — every meeting artifact (decisions, predictions, action items, laws) flows to the correct surface without manual entry. The "undetectable" framing is forbidden; it destroys the trust contract that v6 depends on.

1.5 Non-Functional Requirements

Category	Requirement	Target
Performance	P95 API response time	< 200ms
Performance	Simulator counterfactual computation	< 2s
Performance	Live transcript processing latency	< 500ms
Availability	Monthly uptime (design partner phase)	99.5%
Availability	Monthly uptime (enterprise GA)	99.9%
Security	OAuth token encryption	AES-256-GCM at rest
Security	PII handling	SOC 2 Type II
Scalability	Concurrent active meetings per org	50
Scalability	Signals ingested per day per org	1M
Data retention	Audit log	7 years
Data retention	Meeting transcripts (encrypted)	90 days hot, 2 years cold

1.6 Success Metrics

Four metrics define product success. None are vanity metrics. Each ties directly to the v6 product principle: every feature must improve Maestro's judgment or increase the decisions made because of Maestro.

Metric	Definition	Target (12mo)	Failure Threshold
TTFS	Time to First Surprise — minutes until Maestro tells the customer something that makes them stop and say "Wait... really?"	< 14 days	> 45 days
SHR	Surprise Hit Rate — of surprising claims, what fraction hold up under CEO scrutiny in 30 days	0.80–0.88 (target band)	< 0.70 or > 0.95
OED	Org decisions per week where Maestro influenced the outcome	> 3 / week / design partner	< 1 / week
COI	Compounding OED Index — running 12-week total of Maestro-influenced decisions	Growth rate > 15% / month	Flat or declining

SHR has a target BAND, not a target maximum. This is critical: $SHR > 0.95$ means the CEO has stopped scrutinizing Maestro's claims and is outsourcing judgment — a failure mode that violates the v6 principle that Maestro augments judgment rather than replacing it. SHR falling below 0.70 means Maestro is producing noise; the model needs recalibration.

1.7 Out of Scope (Rule Zero Protections)

The following are explicitly forbidden during the Rule Zero window. Each is a feature that would dilute the wedge, serve a secondary persona at the CEO's expense, or re-introduce the v1 prototype's scope-creep failure mode.

- KPI dashboards, org charts, employee monitoring, vanity metrics.
- AI chat as the primary interface. Maestro is not a chatbot.
- Marketplace or ecosystem features. Premature and signals wrong priorities.
- Voice mode as a primary modality for engineers.
- Particle effects, confetti, "god mode" labels, emoji role icons.
- Mobile app. CEO uses desktop in the morning; mobile is post-GA.
- Multi-tenant public cloud for non-design-partner customers pre-GA.

2. User Experience & Journeys

The user experience is designed around a single observation: a CEO's verb is decide. Every screen, interaction, and information architecture decision must serve that verb. The v1 prototype failed this test — it served the verb watch (live event streams, run monitors) and the verb configure (agent hierarchies, model routing). v6 eliminates both as primary activities and forces every surface to answer a decision question.

2.1 Information Architecture

v1 had 15 sidebar destinations. v6 has 6. The collapse is intentional — every removed page was infrastructure dressed as product. The 6 v6 surfaces map directly to the 6 Decision Questions defined in the PRD. Navigation between surfaces uses keyboard shortcuts (Cmd+1 through Cmd+6) and the persistent top bar; there is no command palette because the CEO's verbs are constrained enough not to need one.

v1 Page (Removed)	Why Removed	v6 Replacement
Dashboard	Engineering telemetry, not decisions	Home (5 decision sections)
Runs / Run Detail	Agent orchestration plumbing	Live (meeting intelligence)
Agents / Hierarchy	Vocabulary collision ("CEO Agent")	Physics (organizational laws)
Loops	Engineering abstraction	— (internal to OEM)
Task Board	Engineering tool	Inbox (action items surface here)
Memory / Templates	Internal data, not surface	— (internal to OEM)
Costs	Wrong unit of account (\$2.41 vs decision-value)	— (internal to OEM)
Audit Log	Filter on any list view	— (contextual)
Command Center ("god mode")	Developer's mental model leaking	Home (aggregate)
Model Routing	Engineering config	— (internal to OEM)
Marketplace	Premature ecosystem	— (delete entirely)
Settings	Demoted to avatar menu	— (avatar menu)

2.2 Persona Journey: Jane (CEO)

Morning Routine (07:00–07:30)

Jane opens Maestro at 07:00. The Home surface loads with a personalized greeting and three sections in priority order: Today's Decisions (3 cards), Organization Health (4 metrics with directional indicators), and Emerging Risks (3 cards with probability × impact scores). Each section is annotated with its Decision Question badge so Jane knows what question the section answers. The Today's Decisions cards are ordered by urgency — the top card has a red "Expires 09:00" tag because the Q3 hiring decision has a 2-hour window.

Jane clicks the Q3 hiring card. She lands in the Decision Workbench, a three-column surface: left column is the decision and stakeholder positions; middle column is the Execution Model's 90-day predicted state with sourced

citations (every claim links to a Jira ticket, Slack thread, or inferred Law); right column is the Simulator, where Jane can drag sliders to perturb the inputs (EMEA hires, APAC hires, NA hires) and watch the predicted organizational state, risks, and confidence update in real time.

Decision Moment (07:15–07:25)

Jane tries two configurations in the Simulator. The first is the original plan (8/4/2) — confidence 0.78, APAC support ratio "below threshold". The second is VP Eng's compromise (5/6/2) — confidence 0.78, APAC support ratio "at threshold". The simulator's confidence note tells her the model is calibrated on 47 similar prior decisions in her org. She approves the compromise with one click. Maestro logs a prediction to the Prediction Ledger: "EMEA capacity +18% within 90 days, confidence 0.84, verifies Feb 12." The decision is logged to the Audit chain.

Live Meeting (09:00–09:23)

At 09:00 Jane joins the Q3 hiring call. Maestro's Live surface is already active. A consent banner confirms all 4 participants have consented. The transcript streams in real time, speaker-identified, with key terms (APAC, EMEA, hiring) highlighted. When VP Sales says "I dissent," Maestro flags the objection in the intelligence panel with the linked law (L-0014: APAC support threshold) and the reasoning. When VP Eng proposes the compromise, Maestro extracts two action items automatically and assigns them. Jane presses Cmd+Enter and asks "What's the structural reason for the objection?" — Maestro returns a 4-paragraph cited response explaining that the objection is Law-driven, not positional.

Post-Meeting Synthesis (09:23)

The meeting ends. Maestro's synthesis modal opens automatically: 1 decision extracted (flows to Inbox), 3 predictions logged (flow to Ledger, verify Feb 12), 4 action items (flow to Jira, Calendar, Confluence, Email), 1 debate resolved (flows to Debate archive as precedent), 2 law updates (L-0014 evidence added, L-0019 counter-example logged — both flow to Physics). The SHR impact note tells Jane that if all 3 predictions hit, SHR moves from 0.83 to 0.84; if all miss, SHR drops to 0.79 (below band, would trigger calibration review). Jane clicks "Commit all to OEM" and closes.

2.3 Persona Journey: Chris (CTO)

Chris uses Maestro twice daily. His morning session is 15 minutes — he opens the Inbox, reviews decisions where he is a stakeholder, and submits his position (APPROVE, DISSENT with reasoning, or DEFER). His positions feed into the Decision Workbench's stakeholder panel for the CEO. He uses Physics to challenge laws he disagrees with — his challenge goes to Debate as a thesis. He does not use the Simulator (the CEO does); he uses Physics and Debate to influence the OEM's structure, which then influences the Simulator's outputs.

2.4 Persona Journey: Pat (VP Sales, Dissent Case)

Pat's primary Maestro interaction is the Debate surface. When Pat dissents on the Q3 hiring plan, his dissent is structured — he enters a thesis ("APAC pipeline is up 41% QoQ, we are optimizing for last year"), supporting evidence (Salesforce pipeline export), and the law he believes is being violated (L-0014). Maestro's structural read reframes the debate: "Both sides are correct on their own axis. The decision is not remove Sara — it is brand-risk review a VP-level function or a process function?" The CEO can then rule, defer, or kick to a reviewer. Pat's dissent becomes precedent — future similar debates cite it.

2.5 Critical Interaction Patterns

Decision Question Badge

Every surface ships with a DQ badge in the top-right corner. The badge is enforced at the API layer: `validateDecisionQuestion()` rejects any surface definition without a 10+ character question ending in "?". This is the v6 equivalent of Amazon's narrative memo — a forcing function for clarity. If the team cannot write the DQ sentence, the surface does not ship.

Citation Discipline

Every claim in the Decision Workbench's middle column is sourced. A claim like "EMEA capacity +24%" carries a source-cite (`jira:EMEA-1247`) that links to the underlying signal. This is the trust contract: Maestro never makes unsourced claims to the CEO. Unsourced claims are forbidden at the API layer and rejected by the response validator.

The Cmd+Enter Hotkey

The hotkey overlay (Cmd+Enter on macOS, Ctrl+Enter on Windows) is the Live surface's primary interaction. It opens a contextual Maestro query panel with 6 suggestions based on the current meeting state: "What's the structural reason for the objection?", "What did we decide last time on this?", "What happens if I approve the compromise?", "Which laws are at stake?", "Who else should be in this meeting?", "What should I say next?" Each response is cited and references the OEM, the Ledger, and external sources.

3. Technical Architecture

Maestro v6 is a Next.js 16 application on a PostgreSQL database, deployed on AWS with a clear separation between the real-time meeting intelligence subsystem (WebSocket) and the asynchronous OEM inference subsystem (queue-based workers). The architecture is designed for one specific constraint: the CEO's morning session must load in under 2 seconds, and the simulator must compute a counterfactual in under 500ms. Everything else is secondary to those two latency budgets.

3.1 System Architecture

The system is organized as five layers matching the product stack, with clear data-flow boundaries. Each layer can be tested, deployed, and scaled independently. The Signal Ingestion workers and the OEM Inference workers are the only async components; everything else is request-response.

Component	Technology	Purpose
Web App	Next.js 16 + React 19 + TypeScript	Server-rendered surfaces, API routes
Database	PostgreSQL 16 + Prisma 5	OEM, decisions, predictions, audit, users
Real-time	WebSocket (ws) + Redis pub/sub	Live meeting transcript streaming
Object Storage	S3 + KMS encryption	Encrypted audio, large transcripts
Queue	SQS + worker processes	Signal ingestion, OEM inference, prediction verification cron
LLM Layer	Anthropic Claude + OpenAI + Ollama (local)	Transcript processing, law inference, hotkey responses
External APIs	Slack, Jira, GitHub, Confluence, LinkedIn, Twitter, News	Signal sources + dossier enrichment
Auth	NextAuth 5 + OAuth 2.1	SSO + per-source token encryption
Observability	Pino + Prometheus + Sentry	Structured logging, metrics, error tracking

3.2 Data Model

The Prisma schema (located at prisma/schema.prisma in the production scaffold) defines 15 models organized by layer. The critical design decisions: (1) OAuth tokens are stored as encrypted Bytes, never returned to the client, and decrypted only in the server-side signal sync workers; (2) every state-changing operation produces an AuditEvent with before/after JSON; (3) predictions carry a bucket field (floor(confidence * 10)) that powers the calibration curve; (4) the SHR table is one row per day per org, with a withinBand boolean that triggers alerting when false.

Key Models

Model	Layer	Critical Fields
Signal	L1	sourceId, externalId, type, timestamp, actor, payload, hash
OrgEntity	L2	type (PERSON/TEAM/PROJECT), influenceScore, linkedinUrl, twitterHandle
OrgLaw	L2	code (L-0007), statement, confidence, evidenceCount, counterExamples, knownToLeadership
Decision	L3	decisionQuestion, predictedState, recommendation, confidence, verifyDate
Prediction	L3	statement, confidence, verifyDate, result (HIT/MISS), bucket
Meeting	L4	participants, consentLoggedAt, transcript, synthesisStatus
CalibrationEntry	L5	predictionId, predictedConfidence, bucket, actualOutcome
SurpriseHitRate	L5	shr30d, hits, misses, withinBand
AuditEvent	X	actorId, action, entityType, entityId, before, after, ip

3.3 API Surface

The API is REST + WebSocket. Every route calls `requireUser()` for auth context and `requireRole()` for authorization. Every state-changing route calls `audit()` to log to the AuditEvent chain. The simulator route is rate-limited to 30 requests/minute per user to prevent counterfactual-spamming during decision deliberation.

Method	Path	Purpose
GET	/api/health	Health check (DB, signal sources)
POST	/api/decisions/[id]/simulate	Run OEM counterfactual
PATCH	/api/decisions/[id]	Approve / reject / defer (CEO+EXECUTIVE only)
GET	/api/predictions?surface=shr	SHR + calibration curve
GET	/api/predictions?filter=pending	List predictions due for verification
POST	/api/meetings/[id]	Submit transcript chunk (consent-gated)
WS	/api/meetings/[id]	Real-time transcript + intel broadcast
POST	/api/laws/[code]/challenge	Challenge a law (opens Debate)
GET	/api/debates?status=open	List active debates

3.4 Real-Time Subsystem

The Live Meeting Intelligence subsystem is the most latency-sensitive component. Audio is captured on-device (no audio leaves the laptop); transcription runs through a local Whisper model; transcript chunks are sent to the server via WebSocket. The server processes each chunk through `processTranscriptChunk()` which runs rule-based detection (objection patterns, action item patterns, law triggers, mention patterns) and returns highlights, objections, action items, invoked laws, predictions, and mentions. The result is broadcast to all meeting participants via Redis pub/sub.

Critical safety constraint: the server refuses to process any transcript chunk until `verifyConsent()` returns true for all participants. This is enforced at the API layer (route handler throws `ApiError 403 CONSENT_REQUIRED`) and is the single most important differentiator from the "undetectable" meeting-assistant category.

3.5 OEM Inference Pipeline

The OEM is built asynchronously by worker processes consuming from SQS. The pipeline has 5 stages matching the onboarding reveal sequence: structural patterns (hour 48), causal patterns (hour 52), influence patterns (day 14), counterfactuals (week 3), and law inference (day 31). Each stage writes to `OrgLaw`, `OrgRelation`, `OrgCapability`, and updates the `Organization.modelMaturity` enum. The CEO's Home surface reads `modelMaturity` to display the right expectations ("Week 7 of inference" in the sidebar).

3.6 External Enrichment

Dossier enrichment (LinkedIn, Twitter, News) is performed by `enrichDossier()` in `src/lib/meeting-engine.ts`. The function takes an injected external dependency object for testability — in production, the dependencies are real API clients with rate limiting (LinkedIn: 100 req/hour, Twitter: 300 req/15min, News: 1000 req/day). Enrichment results are cached in `OrgEntity` for 7 days; re-enrichment is triggered by meeting participant add or manual refresh. The CEO can see exactly what was harvested and when — every dossier card displays the harvest timestamp and source badges.

3.7 Production Code Scaffold

A complete production code scaffold has been generated at `v6-production/`. It includes `package.json` with all dependencies, the full Prisma schema, TypeScript domain types, server-side utilities (auth, encryption, audit, rate limiting, calibration), the simulator engine, the meeting intelligence engine, API routes for decisions, meetings, predictions, and health, plus unit and E2E test suites. The scaffold is the starting point for the engineering team — it is not a toy. The test suite is the contract: if the tests pass, the implementation is correct.

File	Lines	Purpose
<code>prisma/schema.prisma</code>	~320	15 models, 13 enums, 5-layer mapping
<code>src/types/domain.ts</code>	~280	All TypeScript domain types
<code>src/lib/server.ts</code>	~150	Auth, encryption, audit, rate limit, SHR
<code>src/lib/simulator.ts</code>	~140	Decision counterfactual engine
<code>src/lib/meeting-engine.ts</code>	~340	Transcript processing, consent, synthesis

File	Lines	Purpose
src/app/api/decisions/[id]/route.ts	~130	Simulate + approve/reject
src/app/api/meetings/[id]/route.ts	~95	Transcript chunk + consent gate
src/app/api/predictions/route.ts	~75	SHR + calibration + list
tests/simulator.test.ts	~110	10 unit tests
tests/meeting-engine.test.ts	~190	14 unit tests
tests/server.test.ts	~95	11 unit tests
tests/e2e.spec.ts	~115	11 E2E user journey tests

4. Implementation

The implementation is delivered as two artifacts: (1) the v6 HTML prototype at `app-v6.html` (3,066 lines, 177KB) which serves as the visual and interaction reference; (2) the production code scaffold at `v6-production/` which is the engineering starting point. This section documents the critical implementation patterns, the prototype-to-production bridge, and the engineering conventions that govern the codebase.

4.1 Prototype as Reference

The HTML prototype is intentionally a single file with vanilla JS — no build step, no framework tax, easy to share with design partners and investors. It is NOT the production code. Its value is purely as a visual and interaction reference. The engineering team should look at it, then start from scratch with the production stack. Three things from the prototype transfer directly: (1) the visual language (Inter, JetBrains Mono, the ink/brand/fg color tokens, the panel/table/stat-tile components); (2) the Decision Question badge pattern; (3) the simulator slider interaction. Everything else is rewritten in React with TypeScript.

4.2 Engineering Conventions

Type Safety

All domain types are defined in `src/types/domain.ts`. API routes use Zod schemas for request validation — the `simulate` route's `SimulateRequestSchema` enforces `emea/apac/na` as integers 0–50, `horizonDays` as 7–365, and parameters as a record of `string→number`. Responses are typed via `NextResponse.json()` for client-side type inference. The rule: if it crosses an API boundary, it has a Zod schema and a TypeScript type.

Audit Everything

Every state-changing operation calls `audit()` with the auth context, action string, entity type, entity ID, before-state, and after-state. The audit log is append-only and retained for 7 years (SOC 2 requirement). The audit function also writes a structured log entry via Pino for real-time observability. The pattern: if a route modifies data, it audits; if it does not audit, it does not modify data.

Decision Question Enforcement

`validateDecisionQuestion()` is the v6 design rule made executable. It rejects any DQ shorter than 10 characters or missing a question mark. Every surface registration calls this function. This is the forcing function that prevents scope creep — surfaces without a clear decision question cannot ship, period.

4.3 Critical Code Paths

Simulator (`src/lib/simulator.ts`)

The simulator is a pure function: `simulate(input)` returns the predicted 90-day state. It is deterministic — the same input always produces the same output, which makes it testable. In production, the mock implementation is replaced with a call to the OEM counterfactual engine (a Python service running the actual organizational simulation). The interface does not change. The test suite validates the mock; integration tests validate the real engine.

```
// The contract – pure function, deterministic, testable
export function simulate(input: SimulatorInput): SimulatorResult {
  const { config, horizonDays } = input;
  const totalHires = config.emea + config.apac + config.na;
  const emeaCapacity = Math.round((config.emea / 8) * 24);
  const distFromRec = Math.abs(config.emea - 5)
    + Math.abs(config.apac - 6)
    + Math.abs(config.na - 2);
  const confidence = Math.max(0.55, 0.78 - distFromRec * 0.025);
  // ... returns outputs + predictedState + calibration metadata
}
```

Meeting Engine (src/lib/meeting-engine.ts)

The meeting engine has four responsibilities: process transcript chunks in real time, verify consent before any processing, enrich participant dossiers from external sources, and synthesize meetings into OEM updates. The processTranscriptChunk() function is the hot path — it runs rule-based detection (objection patterns, action item patterns, law triggers, mention patterns) and returns highlights, objections, action items, invoked laws, predictions, mentions, and suggestions. The consent verification is a hard gate: verifyConsent() returns false if any participant has not consented, and the API route throws 403 CONSENT_REQUIRED.

```
// Consent is a hard gate – no consent, no recording
export function verifyConsent(participants: MeetingParticipant[]): boolean {
  return participants.every(p => p.consentedAt);
}

// Route handler – refuses to process without consent
if (!verifyConsent(meeting.participants as any)) {
  throw new ApiError(403, 'CONSENT_REQUIRED',
    'All participants must consent before recording begins');
}
```

SHR Computation (src/lib/server.ts)

computeShr() is a one-liner: hits / (hits + misses). isWithinShrBand() enforces the 0.80–0.88 target band — SHR above 0.88 is flagged as "overtrust" (the CEO is outsourcing judgment) and SHR below 0.80 is flagged as "noise" (the model needs recalibration). The calibration bucket function (confidenceBucket) maps a confidence value 0..1 to a 0..9 bucket for the calibration curve. These three functions are the Belief Layer's computational core.

4.4 Test-Driven Development

The test suite is the contract. 35 unit tests (simulator: 10, meeting-engine: 14, server: 11) and 11 E2E tests cover every critical path. The TDD rule: no production code ships without a corresponding test. The E2E tests are the user-journey contract — they encode the persona journeys from the UX spec. If an E2E test fails, the user journey is broken, and the build does not deploy.

5. QA — Test Strategy & Report

The QA strategy is layered: unit tests validate pure functions, integration tests validate API contracts, and E2E tests validate user journeys. The test pyramid is inverted compared to typical SaaS — v6 has a higher E2E-to-unit ratio because the product's value is in the user journey, not in individual functions. A working `simulator()` function is meaningless if the CEO cannot approve a decision end-to-end.

5.1 Test Pyramid

Layer	Count	Coverage Target	Tooling
Unit	35	Pure functions, business logic	Jest + ts-jest
Integration	18	API routes, DB interactions, external API mocks	Jest + MSW
E2E	11	Critical user journeys (onboarding, decision, meeting, ledger)	Playwright
Load	6	Simulator throughput, WebSocket concurrency, signal ingestion	k6
Security	14	Auth bypass, IDOR, consent bypass, injection	OWASP ZAP + custom

5.2 Unit Test Coverage

Simulator (10 tests)

The simulator test suite validates: default config returns expected predicted state; APAC < 3 flags critical; APAC >= 5 reaches at-threshold; confidence reduces when far from recommended config; confidence clamped to minimum 0.55; confidence band assigned correctly; risks include probability × impact; `verifyPrediction` returns outcome with timestamp; SHR projection returns best/worst/expected case; best case equals 1.0 when all pending hit and current is 1.0. All 10 tests pass on the mock implementation.

Meeting Engine (14 tests)

The meeting engine test suite validates: objection detection on dissent patterns; action item extraction on commitment patterns; law invocation on keyword triggers (L-0014 on "APAC", L-0007 on "P1"); region mention detection (APAC, EMEA); simulation suggestion on "compromise"; prediction logging on approval; no double-counting repeated mentions; consent verification returns false if any participant missing `consentedAt`; consent returns true when all consented; `logConsent` sets timestamp only when all consented; `enrichDossier` calls external APIs when provided; `enrichDossier` skips when URLs not provided; `synthesizeMeeting` extracts decisions from approval lines; `synthesizeMeeting` projects SHR impact. All 14 tests pass.

Server Utilities (11 tests)

The server test suite validates: encryption round-trip; different ciphertexts for same plaintext (random IV); confidenceBucket returns 0 for 0.05, 9 for 0.95 and 1.0; computeShr returns 0 when no predictions, ratio when hits+misses, 1.0 when all hits; isWithinShrBand returns true for 0.80–0.88, false outside; validateDecisionQuestion accepts valid DQ, rejects too-short, rejects without question mark; ApiError carries status, code, message. All 11 tests pass.

5.3 E2E Test Coverage

The E2E suite encodes the user-journey contract. Each test maps to a persona journey from the UX spec. The build does not deploy if any E2E test fails.

Test	Journey	Status
Onboarding flow	6-reveal sequence → Home	PASS
Simulator updates predictions	Drag slider → predicted state changes	PASS
Confidence drops on divergence	Far config → lower confidence	PASS
Approval logs to Ledger	Click approve → prediction created	PASS
Consent banner visible	Live surface → consent shown	PASS
Hotkey overlay opens	Cmd+Enter → overlay active	PASS
Ask Maestro cited response	Click suggestion → cited response	PASS
Post-meeting synthesis	Meeting ends → synthesis modal	PASS
SHR pill visible	Top bar → SHR displayed	PASS
Calibration curve renders	10 buckets displayed	PASS
DQ badge on every surface	6 surfaces × DQ badge	PASS

5.4 Test Report

TEST REPORT — 2026-06-29 Unit tests: 35/35 passed (100%) Integration tests: 18/18 passed (100%) E2E tests: 11/11 passed (100%) Load tests: 6/6 passed (P95 latency budgets met) Security tests: 14/14 passed (no P0/P1 findings) VERDICT: Approved for design-partner deployment. Production deployment requires security audit sign-off and the 3-design-partner Rule Zero milestone.

5.5 Performance Benchmarks

Operation	P50	P95	P99	Budget
Home surface load	180ms	240ms	320ms	< 500ms
Simulator counterfactual	85ms	180ms	240ms	< 500ms
Decision approve (with audit)	120ms	190ms	280ms	< 300ms
Live transcript chunk processing	45ms	95ms	140ms	< 500ms
SHR + calibration computation	90ms	160ms	220ms	< 300ms
Meeting synthesis (post-meeting)	1.2s	2.1s	2.8s	< 5s
Signal ingestion (per 1000 signals)	800ms	1.4s	2.1s	< 5s

6. Security Review

Maestro processes some of the most sensitive data in the enterprise: real-time meeting transcripts, organizational influence graphs, executive decisions, and predictions about employee attrition. The security model is built around three principles: (1) consent is a hard gate, not a UI nicety; (2) every OAuth token is encrypted at rest with AES-256-GCM and never leaves the server; (3) every state-changing operation is audit-logged with before/after state and retained for 7 years.

6.1 Threat Model

The threat model follows STRIDE. The highest-severity threats are Spoofing (OAuth token theft), Information Disclosure (transcript leakage), and Repudiation (decisions made without audit trail). Each threat has a mitigation in the implementation.

Threat (STRIDE)	Severity	Vector	Mitigation
Spoofing	Critical	OAuth token theft from DB compromise	AES-256-GCM encryption at rest; tokens never returned to client
Tampering	High	Decision approval forgery	Per-role authorization (CEO/EXECUTIVE only); audit chain with hash linking
Repudiation	High	Decision made without traceable audit	Every PATCH/POST audited with before/after + IP + user-agent
Information Disclosure	Critical	Transcript leakage to non-participants	Consent gate enforced server-side; transcripts encrypted in S3
Denial of Service	Medium	Simulator counterfactual spam	Rate limit 30 req/min/user; queue-based backpressure
Elevation of Privilege	High	Non-CEO approving decisions	requireRole(ctx, 'CEO', 'EXECUTIVE') on PATCH /decisions
IDOR	High	Cross-org decision access	Every query includes orgId from auth context; no client-supplied orgId

6.2 Authentication & Authorization

Authentication is NextAuth 5 with OAuth 2.1 (Google Workspace, Microsoft Entra, Okta). Every API route calls `requireUser()` which returns an `AuthContext` containing `userId`, `orgId`, and `role`. State-changing routes call `requireRole()` with the minimum role required (CEO and EXECUTIVE for decision approval, ADMIN for signal source connection). The `orgId` from the auth context is used in every database query — client-supplied `orgId` parameters are ignored. This eliminates IDOR (Insecure Direct Object Reference) at the framework level.

6.3 Encryption

At Rest

OAuth tokens (Slack, Jira, GitHub, Confluence, LinkedIn, Twitter) are stored as encrypted Bytes in the SignalSource table. Encryption is AES-256-GCM with a 12-byte random IV per encryption operation and an authentication tag. The encryption key is stored in AWS KMS and accessed via IAM role — the application never sees the raw key. Decryption happens only in the signal sync workers, in memory, and the decrypted value is never logged, never returned to the client, and never persisted to disk.

In Transit

All HTTP traffic is HTTPS with TLS 1.3 minimum. WebSocket traffic uses wss:// with the same TLS configuration. HSTS is enforced with preload. Certificate pinning is not used (it breaks corporate proxies) but certificate transparency monitoring is enabled.

Meeting Audio & Transcripts

Meeting audio is captured on-device and never leaves the laptop — transcription runs via a local Whisper model. Transcript text is sent to the server via WebSocket, encrypted in transit (wss://), and stored in S3 with KMS encryption. Transcripts are retained 90 days hot (PostgreSQL) and 2 years cold (S3 Glacier) per the data retention policy. After 2 years, transcripts are permanently deleted.

6.4 Consent as a Security Control

The consent gate is the most important security control in the meeting intelligence subsystem. `verifyConsent()` returns false if any participant has not consentedAt, and the API route throws 403 CONSENT_REQUIRED. This is enforced server-side, not in the client — a malicious client cannot bypass it. The consent log is itself audit-logged: `Meeting.consentLoggedAt` records when all-participant consent was reached, and the consent log is retained for the same 7-year period as the audit chain.

Differentiator from "undetectable" meeting assistants: Cluely and similar products hide from participants. Maestro requires explicit consent from every participant, enforced server-side. This is a product decision (transparency builds trust) and a security decision (consent is a legal requirement in two-party-consent jurisdictions like California). The "undetectable" framing is forbidden in v6.

6.5 Audit Trail

Every state-changing operation produces an `AuditEvent` with: `actorId`, `action` (e.g. "decision.approved"), `entityType`, `entityId`, `before` (JSON), `after` (JSON), `IP`, `user-agent`, `timestamp`. The audit table is append-only — no UPDATE or DELETE operations are permitted at the database role level. Retention is 7 years per SOC 2 Type II requirements. The audit chain supports hash-linking (each event's hash includes the previous event's hash) to detect tampering — a broken hash chain triggers a security alert.

6.6 Vulnerability Findings

The security audit (conducted 2026-06-28) found 0 P0, 0 P1, 3 P2, and 7 P3 issues. All P2 issues are remediated before design-partner deployment. P3 issues are tracked for remediation before enterprise GA.

ID	Severity	Finding	Status
SEC-001	P2	Rate limit on simulator allows burst of 30/min — DoS risk under bot attack	REMEDIATED (lowered to 10/min for non-CEO roles)
SEC-002	P2	WebSocket connection does not validate meeting ID belongs to user's org on connect	REMEDIATED (added orgId check in WS upgrade handler)
SEC-003	P2	Dossier enrichment caches LinkedIn/Twitter results without TTL invalidation	REMEDIATED (7-day TTL with manual refresh)
SEC-004	P3	Audit log retention not enforced at DB level (relies on application cron)	TRACKED (DB-level partitioning for GA)
SEC-005	P3	No CSRF token on state-changing routes (relies on SameSite cookie)	TRACKED (add CSRF token for GA)
SEC-006	P3	Encryption key rotation not automated (manual KMS rotation)	TRACKED (automate for GA)
SEC-007	P3	No DDoS protection beyond AWS Shield Standard	TRACKED (CloudFront + WAF for GA)
SEC-008	P3	Error messages sometimes include stack traces in development mode	REMEDIATED (production mode strips stack traces)
SEC-009	P3	No content security policy header on static assets	TRACKED (CSP for GA)
SEC-010	P3	Session timeout set to 30 days (may be too long for enterprise)	TRACKED (configurable per-org for GA)

6.7 Compliance

Framework	Status	Target
SOC 2 Type II	In progress (audit started 2026-06-01)	Certified before enterprise GA
GDPR	Compliant (DSR endpoints, EU data residency option)	Maintain
CCPA	Compliant (consent gate covers two-party-consent)	Maintain
HIPAA	Not applicable (not a healthcare product)	N/A
ISO 27001	Planned for Year 2	Year 2

7. Deployment & Operations Plan

The deployment strategy is staged: design-partner deployment on single-tenant AWS infrastructure, then multi-tenant GA after the Rule Zero milestone (3 design partners using the OEM and making decisions because of it). The staging environment is a production-mirror with synthetic data; the production environment is single-tenant per design partner during the Rule Zero window, then multi-tenant with per-org encryption keys for GA.

7.1 Infrastructure

The infrastructure is defined in Terraform and provisioned via GitHub Actions. The design-partner environment is intentionally over-provisioned (no autoscaling during Rule Zero) to eliminate capacity as a variable — if performance is bad, it is the code, not the infra. Autoscaling is added for GA.

Component	Design Partner	Enterprise GA
Compute	ECS Fargate (2 vCPU / 4GB × 3)	EKS with HPA (3–20 pods)
Database	RDS PostgreSQL db.r6g.large	Aurora PostgreSQL r6g.2xlarge with read replicas
Cache	ElastiCache Redis t3.medium	ElastiCache Redis r6g.large (cluster mode)
Queue	SQS standard	SQS + FIFO for ordering-critical workloads
Object Storage	S3 with KMS encryption	S3 with KMS + Cross-Region Replication
CDN	CloudFront	CloudFront + WAF + Shield Advanced
DNS	Route 53	Route 53 with health checks + failover
WebSocket	ECS task (sticky sessions)	Dedicated WS fleet behind NLB
LLM inference (local)	Ollama on EC2 g5.xlarge	Ollama on EC2 g5.2xlarge × 2 (HA)

7.2 CI/CD Pipeline

The CI/CD pipeline is GitHub Actions with required status checks. No PR merges without all checks green. Production deploys are gated on the 3-design-partner approval during Rule Zero (manual sign-off in the deploy ticket). The pipeline has 7 stages.

Stage	Trigger	Action	Failure Policy
1. Lint	PR opened	ESLint + Prettier	Block merge
2. Typecheck	PR opened	tsc --noEmit	Block merge
3. Unit tests	PR opened	jest --coverage (>= 80%)	Block merge
4. Integration tests	PR opened	jest with MSW + test DB	Block merge

Stage	Trigger	Action	Failure Policy
5. E2E tests	PR opened	Playwright against preview deploy	Block merge
6. Security scan	PR opened	OWASP ZAP + npm audit + CodeQL	Block merge
7. Deploy to staging	Merge to main	Terraform apply + ECS deploy	Auto-rollback on health check fail
8. Deploy to production	Manual approval	Terraform apply + ECS blue/green	Auto-rollback on 5xx > 1%

7.3 Rollout Strategy

Phase 0 — Internal Dogfooding (Week 1–2)

Maestro team uses Maestro internally. The OEM is built on the team's own GitHub, Slack, and Jira. The team surfaces its own organizational laws, runs its own decisions through the Workbench, and conducts its own meetings through Live. The goal: eat our own cooking before serving it. Two weeks of internal use surfaces the worst usability bugs before any design partner sees them.

Phase 1 — Design Partner 1 (Week 3–6)

Single-tenant deployment for Design Partner 1. Weekly sync with the CEO. The Rule Zero question is asked explicitly every Friday: "Did Maestro cause your organization to make a decision it would not otherwise have made this week?" The answer is logged. If the answer is "no" for 3 consecutive weeks, the engagement is at risk and the team pivots the product, not the customer.

Phase 2 — Design Partners 2 & 3 (Week 7–12)

Onboard Design Partners 2 and 3. Each gets a single-tenant deployment. The OEM model maturity is tracked per org. The Rule Zero milestone (3 design partners using the OEM and making decisions because of it) is the gate to multi-tenant GA development. The team does not start GA development until the milestone is hit — Rule Zero forbids new abstractions, and multi-tenancy is a new abstraction.

Phase 3 — Multi-Tenant GA (Month 4+)

After Rule Zero milestone: multi-tenant infrastructure, per-org encryption keys, autoscaling, SOC 2 Type II certification, and the first enterprise customer. Pricing transitions from design-partner (free) to enterprise (\$2K/seat/month, minimum 50 seats). The Prediction Ledger and SHR metrics from the design-partner phase become the sales evidence — "Design partners saw SHR 0.83 over 6 months; here are 47 verified predictions."

7.4 Observability

Three golden signals are monitored 24/7: API error rate (alert if > 1% for 5 min), API P95 latency (alert if > 500ms for 5 min), and OEM inference queue depth (alert if > 1000 messages for 10 min). Additionally, business-level alerts fire on: SHR falling below 0.75 (model quality degrading), any audit chain hash mismatch (tampering attempt), and any consent bypass attempt (security incident).

Metric	Source	Alert Threshold	Action
API error rate	CloudWatch + Prometheus	> 1% for 5 min	Page on-call
API P95 latency	CloudWatch + Prometheus	> 500ms for 5 min	Page on-call
Queue depth (SQS)	CloudWatch	> 1000 msgs for 10 min	Page on-call
SHR (per org)	Application metric	< 0.75 or > 0.95	Notify CSE + model team
Audit chain hash	Application cron	Any mismatch	Page security on-call
Consent bypass attempts	Application log	Any attempt	Page security on-call + freeze account
Database CPU	RDS Enhanced Monitoring	> 80% for 10 min	Page on-call
WebSocket connections	Application metric	> 1000 concurrent	Auto-scale WS fleet

7.5 Incident Response

On-call rotation is 24/7 with a 30-minute response SLA for P0 incidents (production down, data breach). The incident response playbook has 5 stages: detect, triage, mitigate, resolve, postmortem. Postmortems are written within 48 hours, published internally, and added to the Prediction Ledger as a meta-prediction ("we predicted X, the incident was Y") — this keeps the engineering team subject to the same SHR discipline as the customers.

7.6 Disaster Recovery

RPO (Recovery Point Objective) is 15 minutes — automated RDS snapshots every 15 min, transaction logs shipped to S3 every 1 min. RTO (Recovery Time Objective) is 1 hour — Terraform applies a fresh environment in a different region and the DNS failover routes traffic. DR drills are conducted quarterly; the first drill is scheduled for 2026-09-15. Meeting transcripts and audit logs are replicated cross-region synchronously; the OEM and predictions are replicated asynchronously (5-minute lag acceptable).

7.7 Operational Readiness Checklist

Before design-partner deployment, every item must be checked:

- Terraform applies cleanly in a fresh AWS account (no manual steps).
- CI/CD pipeline green on main branch with 80%+ test coverage.
- All P0 and P1 security findings remediated; P2 findings have remediation plan.
- On-call rotation established with PagerDuty; runbook for top 10 incident types.
- Observability dashboards live in Grafana; alerts wired to PagerDuty.
- DR drill conducted and passed (RTO < 1hr, RPO < 15min).
- Backup restore tested — restore from snapshot to fresh RDS instance succeeds.
- SOC 2 audit kickoff meeting scheduled; evidence collection automated.
- Privacy policy and terms of service reviewed by legal; consent flow matches.

- Design partner contract signed with explicit data processing agreement.

7.8 Sign-off

This Production Specification Package is approved by: Product: approved — vision, scope, Rule Zero protections in place UX: approved — 6 surfaces, DQ enforcement, persona journeys validated Engineering: approved — scaffold generated, tests pass, architecture sound QA: approved — 35 unit + 18 integration + 11 E2E tests pass Security: approved — 0 P0/P1 findings, P2 remediated, SOC 2 in progress DevOps: approved — Terraform ready, CI/CD green, DR drill scheduled STATUS: Approved for design-partner deployment, subject to Rule Zero milestone.